# Incentivization or expropriation? All ESOPs are not created equal

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**ABSTRACT**

Firms often claim that they adopt Employee Stock Ownership Plans (ESOPs) to incentivize employees, but we find controlling shareholders in Chinese listed firms could also tunnel through ESOPs from 2014 to 2018. Firms with higher salary level are more likely to adopt ESOPs to improve worker incentives. However, ESOP adoption negatively affects the performance of firms with high leverage, intercorporate loans, and separation of ownership and control, resulting in high delisting risk. Their controlling shareholders tend to use earnings management, leveraged ESOPs, and ESOPs with high participation rates to inflate the stock prices and then cash out soon after ESOP adoption announcements, siphoning billions of RMB from other investors. These firms tend to be small and announce ESOP adoption when market sentiment is high. Controlling owners in Chinese listed firms choose this myopic form of tunneling after regulatory tightening in 2005, because they can divert more from new investors than from existing corporate assets and their return from tunneling still far exceeds their return from investing in operations.

Keywords: Earnings management; Employee stock ownership plan; Leverage; Minority shareholder expropriation; Salary; Tunneling

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## 1. Introduction

Firms that initiate Employee Stock Ownership Plans (ESOPs) often claim that their goal is to increase employee productivity and shareholder value, but there is mixed empirical evidence on whether ESOP adoption improves firm performance (Faleye, Mehrotra, and Morck, 2006; Kim and Quimet, 2014; O'Boyle, Patel, and Gonzalez-Mulé, 2016). Previous studies have found the heterogeneity in motives for ESOP adoption, such as employee incentivization, takeover defense, and cash conservation.

Chinese listed firms may adopt ESOPs for heterogeneous reasons, but the concentrated corporate ownership and lack of tax incentive exclude takeover defense and tax benefit as motives for ESOP adoption in China. Anecdotal evidence abounds that employees of Chinese listed firms suffer huge losses in ESOPs. According to company announcements, the controlling shareholder of Kaidi Eco (stock code 000939) guaranteed minimum return to the senior tranche of the company's leveraged ESOP, but eventually refused to pay and the employees lost all of their cash contributions, or 130 million RMB, in June 2017. As another example, the controller of Corun (stock code 600478) sold his shares at 180 million RMB to his employees in June 2015, who lost 61% when sales restrictions for the ESOP expired. The weak investor protection and concentrated ownership in China leads to tunneling as a possible motive for ESOP adoption of certain firms, which is not examined by previous studies. Tunneling is the expropriation of minority shareholders by controlling shareholders, who are also the CEOs in most cases (Johnson et al., 2000; La Porta et al., 1999 and 2000). Empirical evidence has shown severe and pervasive tunneling in emerging markets (Berkman, Cole, and Fu, 2009; Chang, 2003; Cheung et al., 2006; Shleifer and Wolfenzon, 2002).

Atanasov et al. (2010) show that more stringent security laws may increase the valuation of firms, but that controlling shareholders tend to increase cash-flow tunneling after new legal rules are enacted to limit equity tunneling. Although Chinese regulatory changes in 2005 and 2006 reduced blatant tunneling through non-operational funds, or intercorporate loans (Jiang, Rao, and Yue, 2015; Li et al., 2020), the controlling shareholders of listed firms may seek new channels for expropriating minority investors.

To fill the void in literature, we investigate whether controlling shareholders in Chinese listed firms adopt ESOPs to improve employee incentives or to expropriate minority shareholders. We study ESOPs adopted in Chinese listed firms from official approval for ESOPs in 2014 to regulatory changes in 2018, using data from Wind and the Chinese Stock Market Research (CSMAR) database. We conduct difference-in-difference (DID) tests on panel data to examine how ESOP adoption affects productivity and Tobin's Q, and we regress firms' abnormal stock returns and delisting risk on firm characteristics which are average values before ESOP adoption announcements.

First, we find the heterogeneity in motives for ESOP adoption in Chinese listed firms. Firms paying higher salary to non-management employees have an increase in productivity and shareholder value from ESOP adoption, which indicates that they adopt ESOPs to improve worker incentives. Consistent with the findings of Kim and Quimet (2014), our results show that the incentive effect of ESOPs is better when there are not too many employees in a firm. Within firms with high salary levels, the performance of firms with too many employees doesn't improve as much as that of firms with not-too-many employees, likely due to the free-rider problem.

However, productivity and shareholder value deteriorate after ESOP adoption for firms with higher levels of leverage, intercorporate loans, and separation of ownership and control. These firms are also more likely to enter financial distress (being delisted for fraud or accounting losses) within 3 years of ESOP adoption announcement, which suggests that controlling owners in firms with poor outlook are willing to tunnel at the expense of hurting employee productivity, as they can expropriate more from new investors than from existing corporate assets. Similar to Qian and Yeung (2015), we find that high leverage in China may signal tunneling, rather than helping to discipline management. Jiang, Lee, and Yue (2010) reported high levels of other receivables in Chinese listed firms, which are evidence of controllers' tunneling through intercorporate loans. Our findings suggest that firms subjected to previous tunneling through intercorporate loans are at higher risk of tunneling through ESOPs, so tunneling tend to persist within a firm.

Second, we investigate how the controlling shareholders tunnel through ESOPs in China. Our tests on short-term abnormal returns and likelihood of controllers' equity sales show that tunneling controllers use ESOP adoption announcement to inflate the stock price and then cash out in the secondary market soon afterwards. They tend to manage earnings upwards before announcing ESOP adoption, provide loans to employees in setting up ESOPs, and persuade as many employees to participate as possible. Controlling shareholders in firms of such traits are much more likely to cash out on ESOP adoption announcements than others, given the same level of short-term abnormal returns. Opportunistic equity sale cannot explain the significantly deteriorating performance and increasing delisting risk of their firms.

Third, we find answers to why public investors and employees get swindled. The pump-and-dump strategy of controlling owners is unlikely to be useful if firm information is easily accessible. However, controllers in small firms can mislead public investors and manipulate stock prices more easily given their higher information asymmetry. As employees are insiders, they are unlikely to participate in ESOPs when the controlling owners have tunneling motives. However, their judgment can be swayed by market sentiment and controlling shareholders' offer of loans and guarantee of minimum returns. DID results show that ESOP adoption negatively affects productivity when the Chinese listed firm has not-so-numerous employees or the ESOP is announced in a month of high investor sentiment ("numerous" and "high" refer to those in the top quartile). In addition, firms with higher delisting risk are much more likely to announce ESOP adoption when investor sentiment is high. Our results suggest that the effect of tunneling overshadows the free-rider effect in China due to its weak legal institutions, and that market optimism is one of the reasons for why employees neglect firm fundamentals.

Tunneling through ESOPs is economically significant. Investors' holding return from ESOP adoption announcement to lockup expiration is reduced by 11.8% and the total market cap by 81.19 billion RMB relative to control groups, when the firms have a high level of earnings management and intercorporate loans. Investors' holding return is reduced by 5.3% and the total market cap by 82.83 billion RMB relative to control groups when the firms use leveraged ESOPs. Because tunneling tends to persist within a firm, the abnormal reduction in market cap at least partially represents the amount of funds siphoned through ESOPs and other channels by the controlling owners.

Our contributions are threefold. First, we find that controllers in firms of poor outlook may tunnel in myopic or even suicidal ways, as the amount of existing corporate assets that they can divert is less than what they can expropriate from new minority investors in the secondary market. Second, our results indicate the persistence of tunneling within a firm. Controlling shareholders may tunnel in more covert ways after regulatory tightening, as long as their return from tunneling still far exceeds their return from investing in operations. Third, our study contributes to the research on ESOPs. Previous studies have identified management entrenchment as a potential misuse of ESOPs in the United States. We identify tunneling risk as a dark side of ESOPs in a country of weak investor protection and concentrated corporate ownership.

The remainder of this paper is organized as follows. Section 2 presents the background and our opposing hypotheses. Section 3 describes the data and presents our methods. Section 4 presents our main results and discussion. Section 5 presents additional tests and discussion on how the controlling shareholders in Chinese listed firms tunnel through ESOPs. Section 6 reports our robustness tests and Section 7 concludes the paper.

## 2. **Background and hypothesis development**

### 2.1. Unique features of ESOPs in Chinese listed firms

In the United States, ESOPs are typically set up as tax-advantaged pension plans, where the tax benefits are associated with positive announcement effects if the ESOP does not replace another retirement plan (Beatty, 1995). Studies on US-based samples show that ESOPs can effectively deter takeovers (Chaplinsky et al., 1994 and 1998; Rauh, 2006) and may alleviate or exacerbate the agency problem depending on the level of management shareholding (Chang and Mayers, 1992). In China, ESOPs are not tax-advantaged and corporate ownerships are concentrated. In 2018, the ownership proportion of the largest shareholder in Chinese listed firms has a median of 31.4% and a 75th percentile of 43.5%, while that of the second largest shareholder has a median of 9.4% and a 75th percentile of 14.9%. Thus, we can exclude tax or takeover concerns as reasons for ESOP adoption in China. The most likely reason might be the expected improvement in firm performance after ESOP adoption.

However, there is mixed evidence on whether ESOPs improve productivity and firm performance. Some researchers found a positive effect of ESOPs on firm performance (Jones and Kato, 1995; O'Boyle, Patel, and Gonzalez-Mulé, 2016), whereas others reported a negligible one (Blasi, Conte, and Kruse, 1996). Ittner et al. (2003) found that firms in high-growth industries tend to have better stock performance after larger-than-expected equity grants to non-executive and non-sales employees. Firms with ESOPs may even deviate from maximizing shareholder value, especially those establishing ESOPs for takeover defense (Faleye, Mehrotra, and Morck, 2006; Gordon and Pound, 1990). According to Kim and Quimet (2014), small ESOPs in firms with not-too-many employees could benefit productivity, but management tends to use large ESOPs to substitute for cash salary or to thwart takeovers.

China first introduced ESOPs in 1992 together with the corporatization of state-owned enterprises (SOEs), but corruption problems led to the termination of ESOPs in listed firms two years later. New regulatory approval of ESOPs came in 2014 for Chinese listed firms. Because the pre-2014 policies may not be relevant today, we only studied the new ESOPs starting from 2014.

ESOP stocks can be acquired in flexible ways for Chinese listed firms, including from secondary markets, private placements, and large shareholders. The China Securities Regulatory Committee (CSRC) places 1-year sales restrictions on ESOPs if their stocks come from the secondary market. The sales restriction is 3 years for those involving private placements. The average lockup period for our sample is 1.3 years, during which period employees are not allowed to sell their shares in ESOPs. Most ESOPs in Chinese listed firms are constructed from stocks purchased in the secondary market, possibly because private placements require regulatory approval and result in a longer lockup period. In our sample, the average time to set up an ESOP is 91 days, starting from the announcement day.

Most Chinese listed firms fund ESOPs with cash from and loans to employees. 4.7% of our sample firms provided full funding for ESOPs, while 95.3% required employees to buy stocks at market prices, either with cash or with loans from large shareholders or financial institutions. And leveraged ESOPs tend to be used when employees buy stocks in the secondary market or in bulk from large shareholders, but the interest rates for leveraged ESOP funding are not disclosed in company announcements. Although a firm's growth prospects may affect its compensation policies (Smith and Watts, 1992) and growth firms are likely to adopt ESOPs to conserve cash, we found that the degree of cash constraint was unrelated to the likelihood of ESOP adoption or to firm performance after establishing ESOPs (data not shown for the sake of brevity).

Chinese ESOPs are generally smaller than US ESOPs (median of ESOP shares to total shares of 1.4% vs. 6.7%), with about 95% of ESOPs in Chinese listed firms making up less than 5% of total shares outstanding. Although ESOP shares account for a small proportion of total shares in China, the participation rates of some plans are disproportionately large. In our sample, the average number of ESOP participants was 444, the median was 180, and the 90th percentile was 1000; the average participation rate was 21.9% (median 12.6%; 90th percentile 51.2%). There is a large variability in plan participation rates. Because the CSRC permits a maximum of 1000 participants in ESOPs of listed firms, the unusually high participation rates (more than 50%) are concentrated in firms with relatively few employees. Overall, the lack of funding, the use of leverage, and the unusually high participation rates for ESOPs in many Chinese listed firms are unlikely associated with incentivization motives.

## 2.2. Hypotheses

Since almost all firms claim that they adopt ESOPs to incentivize employees, we explicitly test whether ESOPs boost employee productivity and shareholder value for Chinese listed firms：

*Incentivization Hypothesis: If the controlling shareholders adopt ESOPs to incentivize employees, productivity and firm performance improve after ESOP adoption.*

Based on the characteristics of ESOPs in Chinese listed firms, some corporate controllers are likely to use ESOPs for non-incentive purposes. 53.5% of ESOP firms in our sample are subsidiaries of business groups, of which the controlling shareholders generally have higher control rights than cash-flow rights. Claessens et al. (2000, 2002) and Lemmon and Lin (2003) show that firms with higher separation of ownership and control suffer more from insider abuse. The weak legal protection for investors, coupled with the separation of ownership and control, is likely to lead to tunneling for certain Chinese listed firms.

Friedman, Johnson, and Mitton (2003) have a model in which the controlling shareholder chooses to prop up the firm when there is a mild adverse shock and to tunnel otherwise. Literature shows the existence of propping in Chinese listed firms (Jian and Wong, 2010; Peng, Wei, and Yang, 2011), where the controlling shareholder injects assets into the firm to avoid delisting. In the year of ESOP adoption announcement, none of the firms are ST or *ST-designated, i.e., facing immediate risks of delisting from the stock markets, so controlling shareholders of Chinese listed firms are unlikely to use ESOPs for propping.

Because controllers often disguise their tunneling as normal operating or financial transactions, researchers have indirectly estimated tunneling based on the market reaction to related-party transactions (Bae, Kang, and Kim, 2002; Baek, Kang, and Lee, 2006), or earnings shock absorptions within a business group (Bertrand, Mehta, and Mullainathan, 2002). Because Chinese listed firms are unlikely to adopt ESOPs to defend takeovers, the short-run market reaction is likely to be positive due to expectations of productivity gains and positive signaling effect. However, if employees are swindled into an ESOP, their morale is likely to be dampened rather than boosted, resulting in deteriorating performance for firms subjected to tunneling.

The discussion above leads to the opposing hypothesis:

*Tunneling Hypothesis: If the controlling shareholder adopts ESOPs to tunnel, productivity and firm performance deteriorate after ESOP adoption.*

Besides productivity and stock performance, the effect of ESOP adoption on delisting risk is important for understanding the myopic behavior of controlling shareholders with tunneling motives, i.e., tunneling at the expense of hurting employee productivity and long-term profitability. For firms on the verge of being delisted, the amount that the controlling owners can tunnel through related party transactions or loan guarantees is limited, because the quality of corporate assets is poor. Although it is costly for controlling shareholders to tunnel using ESOPs, they would care less about firm operations if their firms are unlikely to continue to be listed, or even to exist for more than a few years.

## 3. Data and methodology

### 3.1. Data

The CSRC issued official approval of ESOPs on June 20, 2014, and the first case of ESOP adoption was announced 20 days later. The CSRC, together with three Ministries, issued the New Asset Management Rules on April 27, 2018; these prohibited excessive leverage, roll forward, and implicit guarantee arrangements in asset management products. These new rules will make it more difficult to set up leveraged ESOPs, to roll forward ESOPs (so that gains or losses are not crystallized), and to guarantee minimum returns to the employees. To avoid the impact of these new rules, we only studied ESOPs announced on or before this date. Therefore, our sample included firms listed on the Shanghai and Shenzhen Stock Exchange that announced ESOP adoption from July 10, 2014, to April 27, 2018. We collected ESOP and financial data from the Wind Financial Terminal, and company ownership data from the CSMAR database. We excluded financial firms, those that announced ESOP adoption within 1 year of listing, and those that issued B- or H-shares, because these might not be comparable to most ESOP-adopting firms in the Chinese market. We also dropped firms that suspended trading for more than 10 out of 30 trading days after ESOP adoption announcements. For companies that established ESOPs in steps or announced ESOP adoption multiple times, we only studied the first announcement for each company. Our sample contained 559 firms.

We used the industry classification provided by Wind Information Co., Ltd., which followed the Global Industry Classification Standard (GICS). Table 1 summarizes the distribution of firms that adopted ESOPs by industry and ownership category. Manufacturing firms accounted for 25.4% of the total sample, followed closely by IT firms. Most firms were private enterprises (86.6%), with SOEs and public enterprises accounting for 9.5% and 3.9% of the total sample, respectively. As defined by Chinese regulators, public enterprises are companies with diluted ownership in China, usually former SOEs partially purchased by private individuals. The ownership concentration of public enterprises is much lower than that of SOEs and private enterprises. Unlike the controlling shareholders of private and public enterprises, those of SOEs are severely restricted from selling their shares in the listed firms. In our sample, there were 26 ESOP adoptions announced in 2014, 207 in 2015, 114 in 2016, 158 in 2017, and 54 in the first 5 months of 2018. The number of adoption announcements peaked in 2015, possibly due to the optimism for the Chinese stock market during this period.

**Table 1**

This table reports the distribution of firms adopting ESOPs by industry and ownership category. The sample consists of 559 companies listed on the Shanghai and Shenzhen Stock Exchange announcing ESOP adoption from July 10, 2014 to April 27, 2018. In ownership category, public enterprises refer to companies with diluted ownership in China, usually former SOEs partially purchased by private individuals.

| Industry | Number of firms | Percentage | Ownership category | Number of firms | Percentage |
|---|---|---|---|---|---|
| Healthcare | 55 | 9.84% | Local SOE | 45 | 8.05% |
| Consumer staples | 39 | 6.98% | Central SOE | 8 | 1.43% |
| Real estate | 16 | 2.86% | Private enterprise | 484 | 86.58% |
| Information technology | 126 | 22.54% | Public enterprise | 22 | 3.94% |
| Manufacturing | 142 | 25.40% | | | |
| Consumer discretionary | 88 | 15.74% | | | |
| Materials | 82 | 14.67% | | | |
| Utilities | 7 | 1.25% | | | |
| Energy | 3 | 0.54% | | | |
| Telecom | 1 | 0.18% | | | |

## 3.2. Methodology

### 3.2.1. Regression models for hypothesis testing

We use Equation (1) and (2) to empirically test whether corporate controllers in China use ESOPs to tunnel or to incentivize employees, and to identify the firm characteristics for differentiating among them. The explanatory variables in Eq. (1) are the DID term PostESOP and its interaction with firm characteristics. The explanatory variables in Eq. (2) include firm characteristics and their interaction terms. In the regressions, we use heteroskedasticity robust standard errors.

$$Firm\ performance_{it} = \mathrm{a} + \mathrm{b}(PostESOP_{it}) + \mathrm{c}(ESOP_i) + \mathrm{d}(Post_t) + \mathrm{e}(control\ variable_{it}) + \varepsilon_{it} \quad (1)$$

$$Firm\ performance_i = \text{a} + \text{b(firm characteristics}_i) + \text{c}(control\ variable_i) + \varepsilon_i \quad (2)$$

We use propensity score matching (PSM) to reduce sample selection bias with regard to ESOP-adopting firms. To find PSM controls, we run pooled logistic regression with book-to-market ratio, leverage, industry, size, and year as independent variables to predict the probability of ESOP adoption by Chinese A-share firms. Because omitted variable bias reduces the accuracy of matching, we find 5 nearest neighbors for each ESOP firm in the same industry and year (nearest-neighbor matching), with replacements. As some PSM firms are reused in the matching process, we include all the PSM firms in the DID tests and cross-sectional regressions, and set the event date of each PSM firm as that of the ESOP firm with the closest propensity score. We perform DID test (1) on productivity and Tobin's Q, and cross-sectional regression (2) on BHAR and the likelihood of delisting. For the logistic regression on delisting risk, the dependent variable is equal to 1 if the firm is designated ST/*ST or (in the process of being) delisted within 3 years of ESOP adoption and to 0 otherwise.

3.2.2. Firm performance after ESOP adoption

To test for the tunneling and incentivization motives, we compare long-term firm performance after ESOP adoption in terms of productivity and shareholder value. To measure productivity, we use Data Envelopment Analysis (DEA) and total factor productivity (TFP). Following Demerjian, Lev, and McVay (2012), we use sales revenue as the output, and net PP&E, net lease, other intangible assets, cost of goods sold, goodwill, capitalized R&D, and SG&A expenses as the inputs for DEA efficiency calculation. We sort the observations into industry-year groups, and calculate the relative efficiency for the companies in each group.

For manufacturing firms, we calculate TFP is the residual from fixed effects regression of Cobb Douglas production function: $\text{Log(Total output)}_{\text{it}} = \text{a}_0 + \text{a}_1\text{Log(Labor)}_{\text{it}} + \text{a}_2\text{Log(Capital)}_{\text{it}}$

$+a_3 \mathrm{Log}(\mathrm{Material})_{it} + \varepsilon_{it}$. Similar to Schoar (2002), we calculate total output as the sum of sales revenue, inventory increment and the increase in inventory devaluation, deflated by PPI. We add the amount of inventory devaluation to better reflect the amount of output produced each year. Because the earliest available book value for PP&E of Chinese listed firms is normally 3 to 5 years before IPO, we proxy capital by the net value of Property, Plant, & Equipment (PP&E), deflated by PPI. Following Levinsohn and Petrin (2003), we proxy Labor by the number of employees in each firm, assuming constant work hours per week per worker. Similar to Giannetti et al. (2015), we calculate raw materials as the sum of the cash outflows for goods and services and the increase in current accounts payable, deflated by PPIRM.

For shareholder value, we use Tobin's Q and buy-and-hold abnormal return (BHAR). Tobin's Q is calculated as the market value of total equity and liabilities divided by the book value of total assets. For BHAR, or long-term abnormal return, the beginning and end of the holding period is the event date and lockup expiration date, respectively. According to CSRC regulations, the lock-up period for ESOPs is at least 12 months and should be 36 months for those involving non-public offerings. Following Lyon, Barber, and Tsai (1999), we matched each sample firm to a reference portfolio with similar size and book-to-market values to alleviate new listing, rebalancing, and skewness bias in BHAR. First, we removed 559 sample firms from non-financial A-shares to create a benchmark group. We ranked the firms in the benchmark group on the basis of size, or the average market capitalization between 2012 and 2014, and formed size quintile portfolios based on these rankings. We further divided each size quintile portfolio into book-to-market quintile portfolios using the average book-to-market ratios between 2012 and 2014. Next, we calculated the average size and book-to-market values for each of 25 reference portfolios. Then, we match each ESOP-adopting sample firm to a reference portfolio with the closest average size and book-to-market value, using each sample firm's average size and book-to-market value in the 3 years before its event date. We calculated BHAR as the difference between each sample firm's holding return and the equal-weighted holding return of its matched control group with the same holding period as the sample firm.

3.2.3. Firm characteristics

Regarding explanatory variables, firm characteristics included CC_separation, Other receivables/assets, Accounting accruals, Non-recur income/profits, Leverage, Independent board, Salary_Nonmgt, and Salary/employee_nonmgt. Control variables included Industry, Year, Ownership category, Book to market, Log(size), Log(sales), R&D/Sales, Ad/Sales, Capital/Labor, and Log(age), all of which are defined in Appendix A.

Control and cash-flow rights separation is the controlling owner's control rights minus cash-flow rights, where control rights are the minimum stake and cash-flow rights are the product of each stake in each chain of ownership, following Claessens et al. (2002). We defined CC_separation as the average control and cash-flow rights separation for 3 years before ESOP adoption announcements. Another variable which is likely to be positively associated with tunneling motive is Leverage, the 3-year average of total liabilities divided by total assets before ESOP adoption announcements. According to Bae, Kang, and Wang (2011), employee-friendly firms are more likely to have lower debt ratios. If a company does not treat employees fairly, it is unlikely to motivate them through ESOPs. In addition, Liu and Tian (2012) argued that entrepreneurs are likely to borrow excess debt for tunneling when the legal protection for investors is weak.

A direct proxy of previous tunneling is Other receivables/assets, the 3-year average of other receivables divided by total assets before the announcement of ESOP adoption. This variable reflects the proportion of corporate assets diverted by controlling shareholders through intercorporate loans. Accounting accruals and Non-recur income/profits are associated with earnings management. Similar to

Liu and Lu (2007), we defined Accounting accruals as total accruals, calculated as the 3-year average of (net income-OCF)/TA prior to ESOP adoption announcements. Non-recur income/profits is the 3-year average of non-recurring income divided by net income prior to ESOP adoption announcements. Bertrand, Mehta, and Mullainathan (2002) reported that controlling shareholders in India used non-operating profit to tunnel income across firms within a business group. As non-recurring items are directly available from the financial statements of Chinese listed firms, we use Non-recur income/profits as a proxy for earnings manipulation and previous tunneling.

We also considered variables which are likely to be positively associated with employee incentivization. Previous findings show that better corporate governance reduces tunneling (Pinkowitz, Stulz, and Williamson, 2006; Bae et al., 2012), so we include corporate governance proxies in the explanatory variables, including CEO duality, audit opinion, and whether the firm is in more economically developed regions of China. Klein (2002) find that more independent boards may better monitor against fraudulent reporting, and Chen et al. (2006) and Liu and Lu (2007) also show that higher board independence may help reduce tunneling in the Chinese stock market. We define the variable Independent board as the 3-year average of the number of independent board members divided by the total number of board members prior to ESOP adoption announcements. A firm's salary level could reflect the importance of employees to the firm, and we subtracted management salary and bonuses from total salary expense, as remuneration to directors could be used for tunneling (Fried, Kamar, and Yafeh, 2020; Urzúa, 2009). Salary_Nonmgt is the 3-year average of non-management employees' salary divided by net income prior to ESOP adoption announcements, where non-management employees exclude board members and upper managers. A similar variable is Salary/employee_nonmgt, the total salary for non-management employees divided by the number of non-management employees.

## 4. Empirical results

### 4.1. How does ESOP adoption affect productivity?

Table 2 reports the results of DID test on DEA efficiency and TFP for the whole sample. The coefficient of PostESOP in Model I and V suggests that ESOP adoption negatively affects the productivity of sample ESOP firms. PostESOP×Other receivables/assets is negative but insignificant. PostESOP×Leverage_lag2 has a negative coefficient, which is statistically significant for DEA efficiency but insignificant for TFP. PostESOP× Salary/employee_nonmgt is positive but insignificant. PostESOP× Salary_Nonmgt is positive at the 5% significance level, which suggests that ESOP adoption improves the TFP of Chinese listed firms with higher salary level.

**Table 2**

Regression (fixed effects) results of firm productivity on firm characteristics using ESOP and PSM firms: The whole sample. The variables are defined in Appendix A. Standard errors are in parenthesis. *, **, and ***denote significance at the 10%, 5%, and 1% levels, respectively.

| Model | I | II | III | IV | V | VI | VII | VIII |
|---|---|---|---|---|---|---|---|---|
| Dependent variable | DEA efficiency | | | | TFP | | | |
| **PostESOP** | **-0.013*** | -0.011 | 0.019 | -0.021** | **-0.136**** | -0.095 | -0.025 | -0.148** |
| | **(0.007)** | (0.008) | (0.014) | (0.011) | **(0.061)** | (0.061) | (0.084) | (0.062) |
| **PostESOP× Other receivables/assets** | | **-0.085** | | | | **-1.822** | | |
| | | **(0.140)** | | | | **(1.556)** | | |
| **PostESOP× Leverage_lag2** | | | **-0.001**** | | | | **-0.002** | |
| | | | **(0.000)** | | | | **(0.002)** | |
| **PostESOP× Salary/employee_nonmgt** | | | | **0.067** | | | | |
| | | | | **(0.073)** | | | | |
| **PostESOP× Salary_Nonmgt** | | | | | | | | **0.006**** |
| | | | | | | | | **(0.002)** |
| Other receivables/assets | | -0.079* | | | | 0.386 | | |
| | | (0.047) | | | | (0.450) | | |
| Leverage_lag2 | | | 0.000 | | | | -0.000 | |
| | | | (0.000) | | | | (0.000) | |
| Salary/employee_nonmgt | | | | -0.005 | | | | |
| | | | | (0.010) | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Salary_Nonmgt | | | | | | | | -0.004** |
| | | | | | | | | (0.002) |
| Log(size) | 0.049*** | 0.048*** | 0.051*** | 0.049*** | -0.132*** | -0.130*** | -0.141*** | -0.135*** |
| | (0.005) | (0.005) | (0.006) | (0.005) | (0.042) | (0.042) | (0.054) | (0.042) |
| Leverage | -0.007 | -0.007 | | -0.007 | 0.029** | 0.028** | 0.024 | 0.029** |
| | (0.005) | (0.005) | | (0.005) | (0.012) | (0.012) | (0.015) | (0.012) |
| R&D/Sales | -0.133*** | -0.133*** | -0.069*** | -0.133*** | -0.607 | -0.617 | -0.206 | -0.623 |
| | (0.030) | (0.030) | (0.024) | (0.030) | (1.279) | (1.281) | (1.412) | (1.279) |
| Ad/Sales | -0.029*** | -0.029*** | -0.338*** | -0.029*** | -10.417** | -10.278** | -11.687*** | -10.431** |
| | (0.009) | (0.008) | (0.063) | (0.009) | (4.153) | (4.184) | (4.489) | (4.147) |
| Capital/Labor | -0.001 | -0.001 | -0.000 | -0.001 | 0.010*** | 0.010*** | 0.009*** | 0.010*** |
| | (0.001) | (0.002) | (0.002) | (0.002) | (0.003) | (0.002) | (0.003) | (0.003) |
| Log(age) | -0.150*** | -0.149*** | -0.122* | -0.149*** | -0.411 | -0.424* | -0.914** | -0.403 |
| | (0.033) | (0.033) | (0.069) | (0.033) | (0.251) | (0.249) | (0.452) | (0.250) |
| Intercept | 1.397*** | 1.397*** | 1.147* | 1.389*** | 2.821*** | 2.831*** | 4.192*** | 2.832*** |
| | (0.281) | (0.281) | (0.600) | (0.280) | (0.671) | (0.672) | (1.202) | (0.669) |
| Industry & year effects controlled | | | | | | | | |
| Observations | 14475 | 14465 | 10336 | 14475 | 3761 | 3755 | 2711 | 3761 |
| R-squared | 0.057 | 0.058 | 0.060 | 0.057 | 0.065 | 0.066 | 0.053 | 0.066 |

The we separate the whole sample into 3 sets of subsamples based on the median of the average Salary/employee_nonmgt, Other receivables/assets, and CC_separation for 3 years before ESOP adoption announcement, respectively. In Table 3, PostESOP is insignificant when Salary/employee_nonmgt is high, but significantly negative when Salary/employee_nonmgt is low. In contrast, Model V and VI show that PostESOP is significantly negative when CC_separation is high, but insignificant when CC_separation is low. PostESOP× Leverage_lag2 is significantly negative when the level of previous intercorporate loans is high, but not significant when low. PostESOP× Salary_Nonmgt is insignificant when CC_separation is high, but significantly positive when low. The corporate governance proxies are not statistically significant, and are not reported.

The results show that ESOP adoption increased the productivity of firms with high salary levels, while reduced the productivity of firms with high levels of intercorporate loans, leverage, and separation between ownership and control. The heterogeneity in motives may explain the lack of productivity boost from ESOP adoption for the whole sample.

**Table 3**

Regression (fixed effects) results of firm productivity on firm characteristics using ESOP and PSM firms: The subsamples. The whole sample is divided into 2 sets of subsamples based on the average Salary/employee_nonmgt, Other receivables/assets, and CC_separation, for 3 years before ESOP adoption announcements. The level “High” refers to being more than the median of the grouping criteria in the second row, and “Low” the opposite. The dependent variable for Model I-VI is DEA efficiency, and the dependent variable for Model VII-VIII is TFP. The variables are defined in Appendix A. Standard errors are in parenthesis. *, **, and ***denote significance at the 10%, 5%, and 1% levels, respectively.

| Model | I | II | III | IV | V | VI | VII | VIII |
|---|---|---|---|---|---|---|---|---|
| Grouping criteria | Salary/employee_nonmgt | | Other receivables/assets | | CC_separation | | | |
| Level | High | Low | High | Low | High | Low | High | Low |
| **PostESOP** | **-0.003** | **-0.018**** | 0.060*** | -0.019 | **-0.024**** | **-0.005** | -0.088 | -0.158*** |
| | **(0.012)** | **(0.009)** | (0.020) | (0.019) | **(0.010)** | **(0.011)** | (0.098) | (0.054) |
| **PostESOP× Leverage_lag2** | | | **-0.002***** | **0.000** | | | | |
| | | | **(0.000)** | **(0.000)** | | | | |
| **PostESOP× Salary_Nonmgt** | | | | | | | **-0.026** | **0.005**** |
| | | | | | | | **(0.017)** | **(0.003)** |
| Leverage_lag2 | | | 0.000 | 0.000 | | | | |
| | | | (0.000) | (0.000) | | | | |
| Salary_Nonmgt | | | | | | | -0.003 | -0.003 |
| | | | | | | | (0.003) | (0.002) |
| Log(size) | 0.054*** | 0.044*** | 0.043*** | 0.059*** | 0.042*** | 0.044*** | -0.116* | -0.104** |
| | (0.007) | (0.007) | (0.008) | (0.009) | (0.007) | (0.007) | (0.059) | (0.050) |
| Leverage | -0.003 | -0.007 | | | -0.004 | -0.014 | 0.026* | 0.054*** |
| | (0.017) | (0.005) | | | (0.003) | (0.017) | (0.014) | (0.011) |
| R&D/Sales | -0.068** | -0.239*** | -0.066 | -0.064** | -0.117*** | -0.100 | 0.496 | -1.793*** |
| | (0.033) | (0.048) | (0.052) | (0.025) | (0.029) | (0.121) | (1.504) | (0.486) |
| Ad/Sales | -0.026*** | -0.045*** | -0.439*** | -0.227*** | -0.290*** | -0.030*** | -12.293*** | 1.626 |
| | (0.007) | (0.007) | (0.101) | (0.081) | (0.087) | (0.009) | (3.466) | (8.299) |
| Capital/Labor | 0.000 | -0.018** | 0.000 | -0.006 | -0.002 | 0.000 | -0.003 | 0.013*** |
| | (0.001) | (0.008) | (0.001) | (0.008) | (0.003) | (0.001) | (0.008) | (0.002) |
| Log(age) | -0.143*** | -0.154*** | -0.107 | -0.125 | -0.144*** | -0.152*** | -0.627 | -0.118 |
| | (0.046) | (0.046) | (0.098) | (0.100) | (0.047) | (0.047) | (0.586) | (0.204) |
| Intercept | 1.306*** | 1.464*** | 1.086 | 1.103 | 1.419*** | 1.455*** | 3.296** | 1.736*** |
| | (0.393) | (0.392) | (0.846) | (0.858) | (0.401) | (0.398) | (1.442) | (0.600) |
| Industry & year effects controlled | | | | | | | | |
| Observations | 7286 | 7189 | 5238 | 5098 | 7300 | 7170 | 1889 | 1872 |
| R-squared | 0.059 | 0.071 | 0.062 | 0.069 | 0.061 | 0.057 | 0.068 | 0.092 |

4.2. Are investors better off after ESOP adoption?

We first test the effect of ESOP adoption on Tobin's Q. We conduct DID tests on 3 sets of subsamples created by dividing the full sample in half based on the median of Salary/employee_nonmgt, Other receivables/assets, and CC_separation before ESOP adoption announcements, respectively, and report the results in Table 4. The results indicate the heterogeneity in motives for ESOP adoption in Chinese listed firms. PostESOP is significantly positive when Salary/employee_nonmgt is high, but insignificant when it is low. In contrast, PostESOP is not statistically significant when CC_separation is high, but significantly positive when it is low. In addition, the coefficient of PostESOP×Leverage_lag2 is significantly negative when Other receivables/assets is high, but not significant when Other receivables/assets is low. The DID tests on Tobin's Q provide consistent causal evidence as the DID tests on productivity. ESOP adoption has a positive influence on shareholder wealth for firms with high salary level, but negative or negligible influence for firms with high levels of intercorporate loans, leverage, and separation between ownership and control.

**Table 4**

Regression (OLS) results of Tobin's Q on firm characteristics using ESOP and PSM firms: The subsamples. The whole sample is divided into 2 sets of subsamples based on the average Salary/employee_nonmgt, Other receivables/assets, and CC_separation, for 3 years before ESOP adoption announcements. The level "High" refers to being more than the median of the grouping criteria in the third row, and "Low" the opposite. The variables are defined in Appendix A. Standard errors are in parenthesis. *, **, and ***denote significance at the 10%, 5%, and 1% levels, respectively.

| Model | I | II | III | IV | V | VI |
|---|---|---|---|---|---|---|
| Grouping criteria | Salary/employee_nonmgt | | Other receivables/assets | | CC_separation | |
| Level | High | Low | High | Low | High | Low |
| **PostESOP** | **0.315**** | **0.141** | 1.966*** | 1.428* | **0.125** | **0.295**** |
| | **(0.148)** | **(0.096)** | (0.314) | (0.839) | **(0.133)** | **(0.115)** |
| **PostESOP× Leverage** | | | **-0.037***** | **-0.035** | | |
| | | | **(0.005)** | **(0.022)** | | |
| Leverage | | | 0.015*** | 0.032 | | |
| | | | (0.002) | (0.023) | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| ESOP | -0.512*** | -0.515*** | -0.692*** | -0.341*** | -0.634*** | -0.324*** |
| | (0.084) | (0.056) | (0.087) | (0.062) | (0.080) | (0.070) |
| Post | -0.432* | -0.320** | -0.538** | -0.344** | -0.378 | -0.461** |
| | (0.243) | (0.138) | (0.271) | (0.136) | (0.238) | (0.192) |
| Log(size) | -0.188*** | -0.138*** | -0.214*** | -0.135* | -0.297*** | -0.101*** |
| | (0.051) | (0.040) | (0.047) | (0.081) | (0.053) | (0.039) |
| Leverage | 1.146*** | 1.747*** | | | 1.576*** | 1.423*** |
| | (0.250) | (0.525) | | | (0.481) | (0.310) |
| R&D/Sales | 1.136 | 0.359 | 1.582*** | 0.674 | 0.735* | 5.187*** |
| | (0.812) | (0.405) | (0.504) | (0.477) | (0.438) | (0.756) |
| Ad/Sales | 0.116 | 4.104*** | 0.731 | 1.903** | 4.078*** | 0.157 |
| | (0.117) | (0.394) | (0.769) | (0.761) | (0.383) | (0.186) |
| Capital/Labor | -0.009** | -0.167** | -0.010* | -0.023** | -0.022** | -0.032*** |
| | (0.004) | (0.078) | (0.006) | (0.012) | (0.010) | (0.012) |
| Log(age) | 0.586*** | 0.660*** | 0.812*** | 0.219 | 0.623*** | 0.318* |
| | (0.195) | (0.158) | (0.203) | (0.171) | (0.172) | (0.185) |
| Intercept | -1.856 | -4.186*** | -4.266*** | -0.163 | -2.180* | -1.114 |
| | (1.481) | (1.260) | (1.623) | (1.277) | (1.248) | (1.423) |
| Industry & year effects controlled | | | | | | |
| Observations | 7288 | 7189 | 7307 | 7170 | 7300 | 7172 |
| R-squared | 0.071 | 0.259 | 0.119 | 0.163 | 0.134 | 0.096 |

Although the DID tests on Tobin's Q adjust for preexisting conditions, they only show the relative change in firm valuation, so we also examine BHAR, or the holding returns of ESOP participants and public investors from ESOP adoption announcements to lockup expiration. In addition, the risk of the firm entering financial distress soon after ESOP adoption could both support the existence of tunneling and explain the myopic behavior of tunneling through ESOPs. Out of the 559 sample firms, 38 are designated ST/*ST, or in the process of being delisted, or already delisted, within 3 years of ESOP adoption. These firms are delisted all due to fraudulent reporting or consecutive years of accounting losses, not due to privatization or takeovers.

Table 5 reports the regression results derived from Eq. (2) for BHAR and delisting risk. Other receivables/assets×Leverage is negatively associated with BHAR, and positively associated with delisting risk, both at the 10% significance level. The coefficient of Salary_Nonmgt is positively significant at the 1% level for BHAR, but insignificant for delisting risk. Accounting accruals×CC_separation is negatively associated with BHAR and positively associated with delisting risk, both at the 5% level.

The cross-sectional findings corroborate DID test results. Firm productivity and shareholder value increase after ESOP adoption when the firm pays higher salary to non-management employees prior to ESOP adoption, which support the incentivization hypothesis for firms with competitive salary levels. However, productivity decreases, market cap shrinks, and delisting risk increases after ESOP adoption when the firm has high levels of leverage, intercorporate loans, and separation of ownership and control, which support the tunneling hypothesis for these firms. Our results also indicate the validity of using other receivables to proxy for previous tunneling in Chinese listed firms, because firms with higher levels of other receivables have worse performance and higher risk of entering financial distress.

**Table 5**

Regression (cross-sectional) results of BHAR and delisting risk on firm characteristics using ESOP firms: The whole sample. The dependent variable for Model IV-VI equals 1 if the firm is designated ST/*ST or delisted within 3 years of ESOP adoption, and 0 otherwise. All financial variables are the 3-year average before ESOP adoption announcements. The variables are defined in Appendix A. Standard errors are in parenthesis. *, **, and ***denote significance at the 10%, 5%, and 1% levels, respectively.

| Model | I | II | III | IV | V | VI |
|---|---|---|---|---|---|---|
| Dependent variable | BHAR | | | Delisting risk | | |
| **Other receivables/assets×Leverage** | **-4.538*** | | | **90.134*** | | |
| | **(2.462)** | | | **(50.203)** | | |
| **Salary_Nonmgt** | | **0.002***** | | | **-0.000** | |
| | | **(0.000)** | | | **(0.002)** | |
| **Accounting accruals×CC_separation** | | | **-0.077**** | | | **0.872**** |
| | | | **(0.035)** | | | **(0.370)** |
| Other receivables/assets | 1.886 | | | -64.942* | | |
| | (1.219) | | | (34.411) | | |
| Leverage | -0.084 | -0.150 | -0.205 | 2.827** | 3.817*** | 3.908*** |
| | (0.187) | (0.171) | (0.177) | (1.209) | (1.134) | (1.149) |
| Accounting accruals | | | 0.204** | | | -7.042*** |
| | | | (0.097) | | | (2.672) |
| CC_separation | | | -0.001 | | | 0.011 |
| | | | (0.003) | | | (0.020) |

| | | | | | | |
|---|---|---|---|---|---|---|
| Log(sales) | 0.022 | 0.023 | 0.027 | -0.298* | -0.285* | -0.292* |
| | (0.027) | (0.028) | (0.029) | (0.159) | (0.164) | (0.175) |
| R&D/Sales | -0.397 | -0.392 | -0.350 | -3.691 | -4.941 | -4.519 |
| | (0.445) | (0.436) | (0.925) | (7.783) | (6.803) | (9.945) |
| Ad/Sales | -0.207 | -0.163 | -0.197 | -1.682 | -1.855 | -1.709 |
| | (0.192) | (0.189) | (0.191) | (2.420) | (2.415) | (2.721) |
| Capital/Labor | 0.047 | 0.053 | 0.042 | 0.432 | 0.481 | 0.291 |
| | (0.043) | (0.043) | (0.044) | (0.344) | (0.353) | (0.332) |
| Log(age) | 0.040 | 0.036 | 0.057 | -0.049 | -0.097 | -0.093 |
| | (0.062) | (0.062) | (0.063) | (0.575) | (0.584) | (0.618) |
| Intercept | -0.532 | -0.489 | -0.629 | -1.338 | -1.707 | -1.541 |
| | (0.629) | (0.639) | (0.638) | (5.066) | (5.013) | (5.251) |
| Industry & year effects controlled | | | | | | |
| Observations | 541 | 541 | 533 | 487 | 487 | 479 |
| R-squared | 0.035 | 0.064 | 0.043 | | | |
| Pseudo $R^2$ | | | | 0.112 | 0.098 | 0.118 |

Although Salary_Nonmgt is insignificantly correlated with delisting risk for the whole sample, Table 6 shows that Accounting accruals×CC_separation is positively associated with delisting risk when Salary_Nonmgt is low, but insignificant when Salary_Nonmgt is high. This confirms that tunneling is more likely to happen in firms with lower salary level before ESOP adoption announcement.

Kim and Quimet (2014) point out that both ESOP size and firm size reflect the motives of ESOP adoption, because cash conservation or takeover defense generally require large ESOPs (those controlling more than 5% of the total shares outstanding) and too many employees bring in the free-rider problem. Because less than 5% of our sample firms adopt large ESOPs, we only separate sample firms into those with numerous and not-so-numerous employees. The former is those in the top quartile in terms of the average number of employees for 3 years before ESOP adoption announcement. In Table 6, Salary_Nonmgt is significantly positively correlated with BHAR when the firm has not-so-numerous employees, but is insignificant when the firm has numerous employees. In DID test for Tobin's Q and TFP, we also find the interaction term PostESOP× Salary_Nonmgt to be significantly positive only when the employees are not-so-numerous (table not shown for the sake of brevity). As shown previously, the level of salary to non-management employees are positively associated with the likelihood of genuine incentivization motives. When firms adopt ESOPs to incentivize employees, there's no noticeable improvement in performance for firms with too many employees, likely due to the free-rider problem.

**Table 6**

Regression (cross-sectional) results of delisting risk and BHAR on firm characteristics using ESOP firms: The subsamples. The whole sample is divided into 2 sets of subsamples based on the average Salary_Nonmgt and the number of employees for 3 years before ESOP adoption announcements. The level "High" refers to being more than the median of Salary_Nonmgt, and "Low" the opposite. Numerous-employee firms are those in the top quartile in terms of the number of employees. The dependent variable for Model I- II equals 1 if the firm is designated ST/*ST or delisted within 3 years of ESOP adoption, and 0 otherwise. All financial variables are the 3-year average before ESOP adoption announcements. The variables are defined in Appendix A. Standard errors are in parenthesis. *, **, and ***denote significance at the 10%, 5%, and 1% levels, respectively.

| Model | I | II | III | IV |
|---|---|---|---|---|
| Dependent variable | Delisting risk | | BHAR | |
| Grouping criteria | Salary_Nonmgt | | Employees | |
| Level | High | Low | Numerous | Not-so-numerous |
| **Accounting accruals×CC_separation** | **0.624** | **0.952*** | | |
| | **(0.705)** | **(0.512)** | | |
| **Salary_Nonmgt** | | | **-0.000** | **0.002***** |
| | | | **(0.002)** | **(0.000)** |
| Accounting accruals | -3.426 | -9.928** | | |
| | (4.774) | (3.887) | | |
| CC_separation | 0.040 | -0.010 | | |
| | (0.031) | (0.033) | | |
| Log(sales) | -0.613** | -0.018 | 0.053 | -0.041** |
| | (0.295) | (0.304) | (0.089) | (0.020) |
| Leverage | 3.499 | 4.710*** | -0.649 | 0.005 |
| | (2.173) | (1.473) | (0.639) | (0.111) |
| R&D/Sales | -32.402 | 10.140 | 7.214 | -0.487 |
| | (20.530) | (14.028) | (6.346) | (0.453) |
| Ad/Sales | 0.621 | -3.697 | -0.398 | -0.311 |
| | (4.059) | (6.719) | (0.482) | (0.207) |
| Capital/Labor | 1.556** | -0.312 | 0.199 | 0.017 |
| | (0.771) | (0.490) | (0.132) | (0.047) |
| Log(age) | 0.347 | -0.172 | 0.235 | -0.045 |
| | (0.859) | (0.915) | (0.148) | (0.068) |
| Intercept | -3.172 | -2.267 | -2.741* | 0.756 |
| | (7.859) | (7.173) | (1.460) | (0.688) |
| Industry & year effects controlled | | | | |
| Observations | 230 | 233 | 135 | 406 |

| | | | | |
|---|---|---|---|---|
| Pseudo R2 | 0.141 | 0.173 | | |
| R-squared | | | 0.116 | 0.118 |

## 5. Additional tests and discussion

### 5.1. How do they tunnel?

#### 5.1.1. Inflating stock prices and cashing out

If the controlling shareholders intend to motivate employees through ESOPs, all else equal, they will be less likely to sell their shares in the firm soon after ESOP adoption announcements due to the expected improvement in productivity and profitability. However, controlling shareholders with tunneling motives are likely to manage earnings upward in previous years, raise the share prices through ESOP adoption announcements, and eventually cash out by selling their shares in the firms soon after ESOP adoption announcements. Due to information asymmetry, other market participants do not expect the large equity sales soon after the ESOP adoption announcement, which helps the controlling shareholder to lock in gains. Out of the 559 controllers of our sample firms, 179 sold their shares in their companies within 3 months of announcing ESOP adoption, which is also the average time to establish an ESOP for our sample. Because the ownership concentration is very high in China, the controlling shareholders can gain by selling their shares without losing their private benefits of control. Given the small ESOP size in Chinese listed firms, the main channel for tunneling owners to cash out is more likely to be equity sales in the secondary market, rather than block sales to ESOPs. Only 1.9% of the ESOPs in our sample acquire all their shares from controlling shareholders. Even if the employees only purchase a small number of shares, ESOP adoption sends out a positive signal to the market and will increase the profit of controlling owners when selling their stocks in the secondary market.

We use the following equations to test price inflation and stock sales as a channel for tunneling through

ESOPs:

$CAR_i$=a + b(firm/ESOP characteristics$_i$) + c($control\ variable_i$) + $\varepsilon_i$ (3)

$Logit(Equity\ sales)_i$=a + b(firm/ESOP characteristics$_i$) + c($control\ variable_i$) + $\varepsilon_i$ (4)

Eq. (3) tests the firm and ESOP characteristics associated with the short-term abnormal return surrounding ESOP adoption announcements from standard event-study approach. The event date is the initial board meeting announcement day, or the next trading day if the announcement is made during non-trading days. All prices are adjusted for stock splits. We use both the market-adjusted and market model to compute the daily abnormal return (AR). For the market model, the estimation window is between 220 and 20 days before the event date. For each method, the cumulative abnormal return (CAR) between dates T1 and T2 is calculated as: $CAR_i(T_1, T_2) = \sum_{t=T_1}^{T_2} AR_{it}$. We test CAR with event windows spanning between the 10 days before and the 30 days after the ESOP adoption announcement day, all of which are significantly positive. We use CAR (-5,5) as the dependent variable for Eq. (3), which has a mean of 2.2% and a t-statistic of 3.7. The positive short-term market reaction could come from expectations of productivity gains or positive signaling effect. For a few sample firms, the controllers guarantee minimum returns to the employees, and the short-run stock price increase is more impressive.

Eq. (4) tests the firm and ESOP characteristics associated with the controller's likelihood of cashing out soon after the ESOP adoption announcement. The dependent variable of Eq. (4) was a dummy, equal to 1 for companies whose controlling shareholders (including their family members) sold stocks in the company within 1 month of ESOP adoption announcements.

ESOP characteristics included ESOP_leverage, ESOP_participation, Log(ESOP_participants), ESOP_Mgt, and ESOP_stock. ESOP_leverage is a dummy variable that equals 1 if the controlling shareholder or financial institutions provide loans to employees for ESOP implementation. ESOP_participation is the number of ESOP participants divided by the number of employees in the year of ESOP adoption. An alternative variable is Log(ESOP_participants), the natural logarithm of the number of ESOP participants. ESOP_Mgt is the percentage of an ESOP owned by management. Zenger and Marshall (2000) found that the incentive intensity of group-based rewards is positively associated with management participation. ESOP_stock is a dummy variable that equals 1 if the controlling shareholder transfers shares to the ESOP via block sales. Because ESOP characteristics are missing for PSM firms and not time-varying, we only use them in the cross-sectional regression for ESOP firms. Because ESOP_stock overlapped in part with the dependent variable of Eq. (4), they are not used together.

Table 7 reports the regression results for our sample from Eq. (3) and (4). These variables are significantly positively associated with CAR: CC_separation×Leverage, Leverage×Non-recur income/profits, Accounting accruals, Private enterprise×ESOP_leverage. CC_separation and Leverage are positively associated with the likelihood of tunneling through ESOPs in Chinese listed firms as shown in Section 4, and they are also positively associated with the short-term abnormal return. Firms with higher levels of earnings management tend to have higher CAR than other ESOP-adopting firms. Investors probably cannot differentiate between controllers with tunneling and incentivization motives and assume all controllers intend to motivate employees by establishing ESOPs. This might explain the positive CAR for these firms, but not their higher CAR than other firms'. More likely, controlling shareholders with tunneling motives use certain tactics to inflate stock prices during ESOP adoption. The interaction between ESOP_leverage and Ownership category results in a non-zero value when the firm is a Private enterprise, because only four public enterprises use leveraged ESOPs and no SOEs use them. The significantly positive correlation between Private enterprise×ESOP_leverage and CAR suggests that controlling owners with tunneling motives may use leveraged ESOPs as a tactic to inflate the stock prices. ESOP_Mgt and ESOP_stock are not statistically significant, and are not reported.

From Model V and VI, ESOP_Leverage×ESOP_participation and Log(ESOP_participants)×Non-recur income/profits are both significantly positively correlated with the likelihood of equity sales. When the Chinese listed firm adopts leveraged ESOPs and ESOPs with more participants, its controlling shareholder are more likely to sell the firm's stocks soon after the ESOP adoption announcement.

**Table 7**

Regression (cross-sectional) results of CAR and likelihood of equity sales on ESOP and firm characteristics using ESOP firms: The whole sample. The dependent variable for Model I-IV is CAR (-5,5). The dependent variable for logistic regression V-VI equals 1 if the controlling shareholder sells the company's stocks within 1 month of the ESOP adoption announcement and 0 otherwise. All financial variables are the 3-year average before ESOP adoption announcements. The variables are defined in Appendix A. Standard errors are in parenthesis. *, **, and ***denote significance at the 10%, 5%, and 1% levels, respectively.

| Model | I | II | III | IV | V | VI |
|---|---|---|---|---|---|---|
| Dependent variable | CAR | | | | Equity sales | |
| **CC_separation×Leverage** | **0.007*** | | | | | |
| | **(0.005)** | | | | | |
| **Leverage×Non-recur income/profits** | | **0.080***** | | | | |
| | | **(0.030)** | | | | |
| **Accounting accruals** | | | **0.003***** | | | |
| | | | **(0.001)** | | | |
| **Private enterprise×ESOP_Leverage** | | | | **0.164***** | | |
| | | | | **(0.049)** | | |
| **ESOP_Leverage×ESOP_participation** | | | | | **1.576**** | |
| | | | | | **(0.803)** | |
| **Log(ESOP_participants)×Non-recur income/profits** | | | | | | **0.513*** |
| | | | | | | **(0.305)** |
| CC_separation | -0.002 | | | | | |
| | (0.002) | | | | | |
| Non-recur income/profits | | -0.031** | | | | -3.361** |
| | | (0.015) | | | | (1.707) |
| ESOP_Leverage | | | | -0.159*** | -0.867** | |
| | | | | (0.047) | (0.399) | |
| Ownership category (ref.= Local SOE) | | | | | | |
| Central SOE | | | | 0.067 | | |
| | | | | (0.082) | | |
| Private enterprise | | | | 0.037 | | |
| | | | | (0.025) | | |
| Public enterprise | | | | 0.114** | | |
| | | | | (0.045) | | |
| ESOP_participation | | | | | 0.309 | |
| | | | | | (0.338) | |
| Log(ESOP_participants) | | | | | | -0.023 |
| | | | | | | (0.125) |

| | | | | | | |
|---|---|---|---|---|---|---|
| Salary_Nonmgt | 0.000*** | 0.000*** | 0.000*** | 0.000 | -0.002 | -0.002 |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.001) | (0.002) |
| Log(size) | -0.019** | -0.017** | -0.037*** | -0.019** | 0.398** | 0.257 |
| | (0.008) | (0.008) | (0.011) | (0.008) | (0.184) | (0.205) |
| Leverage | 0.139 | 0.090 | -0.007 | 0.049 | -0.783 | -0.361 |
| | (0.148) | (0.155) | (0.036) | (0.032) | (0.800) | (0.815) |
| R&D/Sales | -0.664 | -0.376 | -0.342 | -0.491** | 1.766 | 1.814 |
| | (0.586) | (0.244) | (0.236) | (0.247) | (3.736) | (4.100) |
| Ad/Sales | 0.155** | 0.135** | 0.171** | 0.164** | 1.916 | 1.600 |
| | (0.067) | (0.067) | (0.067) | (0.069) | (1.281) | (1.512) |
| Capital/Labor | 0.022 | 0.015 | 0.021 | 0.029* | -0.554 | -0.508 |
| | (0.015) | (0.013) | (0.015) | (0.015) | (0.420) | (0.435) |
| Log(age) | 0.008 | 0.011 | 0.009 | 0.012 | -0.778* | -0.643 |
| | (0.019) | (0.019) | (0.019) | (0.019) | (0.406) | (0.409) |
| Intercept | 0.057 | 0.009 | 0.076 | -0.033 | 2.519 | 2.691 |
| | (0.187) | (0.180) | (0.180) | (0.188) | (3.685) | (3.766) |
| Industry & year effects controlled | | | | | | |
| Observations | 533 | 541 | 541 | 540 | 488 | 489 |
| R-squared | 0.082 | 0.091 | 0.082 | 0.092 | | |
| Pseudo $R^2$ | | | | | 0.058 | 0.052 |

Table 8 reports the regression results derived from Eq. (4) for the subsamples. CAR is positively associated with controlling shareholders' cashing out at the 1% significance level when Accounting accruals, ESOP_participation, or Other receivables/assets is high, but is not significant when Accounting accruals, ESOP_participation, or Other receivables/assets is low. Accounting accruals×CAR is significantly positive at the 1% level when Leverage is high, but not significant when Leverage is low. The results support the heterogeneity in motives for ESOP adoption in China, because controlling shareholders are not selling their stocks in their firms opportunistically (when CAR is high). High levels of leverage and intercorporate loans are associated with higher short-term abnormal return, greater likelihood of controllers' immediate equity sales, deteriorating firm performance and increasing delisting risk. High levels of intercorporate loans, which are associated with previous tunneling through non-operational funds, precede tunneling through ESOPs. Continuous tunneling in such firms could cause declining firm performance after ESOP adoption. After cashing out, corporate controllers are likely to abandon the facade of robust growth and may even engage in other forms of tunneling.

**Table 8**

Logistic regression (cross-sectional) results of likelihood of equity sales on firm characteristics using ESOP firms: The subsamples. The whole sample is divided into 2 sets of subsamples based on the average Accounting accruals, ESOP_participation, Other receivables/assets, and Leverage for 3 years before ESOP adoption announcements. The level "High" refers to being more than the median of the grouping criteria in the second row, and "Low" the opposite. The dependent variable equals 1 if the controlling shareholder sells the company's stocks within 1 month of the ESOP adoption announcement and 0 otherwise. All financial variables are the 3-year average before ESOP adoption announcements. CAR is CAR (-10,30). The variables are defined in Appendix A. Standard errors are in parenthesis. *, **, and ***denote significance at the 10%, 5%, and 1% levels, respectively.

| Model | I | II | III | IV | V | VI | VII | VIII |
|---|---|---|---|---|---|---|---|---|
| Grouping criteria | Accounting accruals | | ESOP_participation | | Other receivables/assets | | Leverage | |
| Level | High | Low | High | Low | High | Low | High | Low |
| **CAR** | **2.867*****  | **0.459** | **3.712***** | **0.246** | **2.406***** | **0.523** | 2.652*** | 2.010** |
| | **(1.022)** | **(1.056)** | **(1.019)** | **(0.984)** | **(0.869)** | **(1.197)** | (0.975) | (0.828) |
| **Accounting accruals×CAR** | | | | | | | **67.235***** | **10.146** |
| | | | | | | | **(16.980)** | **(19.336)** |
| Accounting accruals | | | | | | | -2.269 | -2.473 |
| | | | | | | | (3.146) | (3.845) |
| Salary_Nonmgt | 0.009 | -0.014 | -0.005* | 0.012 | -0.002 | -0.002 | 0.015 | -0.095 |
| | (0.011) | (0.024) | (0.003) | (0.025) | (0.014) | (0.001) | (0.010) | (0.114) |
| Leverage | -2.146* | -0.049 | -0.340 | -0.948 | -1.115 | -0.417 | -0.956 | -1.591 |
| | (1.191) | (1.075) | (1.079) | (1.171) | (1.213) | (1.114) | (2.046) | (1.971) |
| Log(size) | 0.322 | 0.545** | 0.583** | 0.276 | 0.572** | 0.308 | 0.547** | 0.427* |
| | (0.262) | (0.253) | (0.260) | (0.282) | (0.290) | (0.250) | (0.277) | (0.248) |
| R&D/Sales | 8.317*** | -8.899 | 8.488*** | -27.049 | 4.925 | -13.377 | 15.196*** | -22.058** |
| | (3.221) | (9.300) | (2.707) | (24.922) | (3.802) | (16.670) | (4.378) | (10.191) |
| Ad/Sales | -0.273 | 1.899 | -0.133 | 5.788*** | 1.208 | 1.561 | -1.760 | 1.887 |
| | (2.605) | (1.819) | (1.609) | (2.079) | (2.061) | (1.565) | (2.216) | (1.851) |
| Capital/Labor | -0.430 | -0.629 | -1.059* | 0.426 | -0.461 | -0.395 | -0.216 | -1.581* |
| | (0.572) | (0.588) | (0.603) | (0.544) | (0.582) | (0.601) | (0.554) | (0.844) |
| Log(age) | -1.333* | -0.232 | -1.119** | -0.803 | -0.875 | -0.405 | -0.874 | -0.805* |
| | (0.725) | (0.505) | (0.543) | (0.777) | (0.662) | (0.547) | (0.749) | (0.489) |
| Intercept | 7.592 | -3.633 | 3.234 | 3.585 | 1.694 | -0.732 | -0.451 | 3.601 |
| | (6.058) | (5.108) | (4.655) | (7.315) | (5.009) | (5.728) | (6.843) | (4.580) |
| Industry & year effects controlled | | | | | | | | |
| Observations | 254 | 262 | 264 | 252 | 237 | 259 | 247 | 269 |
| Pseudo $R^2$ | 0.122 | 0.049 | 0.127 | 0.084 | 0.118 | 0.031 | 0.161 | 0.118 |

Then we analyze the influence of ESOP adoption on firm performance when earnings management and ESOP participation rate are high, and when leveraged ESOP are used. Some ESOP characteristics are missing for a few firms, so we do not use them in delisting risk regression. In Table 9, Accounting accruals, Private enterprise×ESOP_leverage, and ESOP_participation× CC_separation are all significantly negatively associated with BHAR. The DID test for DEA efficiency shows that PostESOP×Accounting accruals_lag2 is negative but insignificant, but the DID tests for TFP and Tobin's Q show that PostESOP×Accounting accruals is significantly negative.

The results support that tunneling controllers manage earnings upward and cash out soon after announcing ESOP adoption. By managing earnings upwards in prior years, controlling shareholders raise market expectations for future growth and, consequently, the share price. The inflated stock prices are further raised by ESOP adoption announcements, which ultimately bolster the controlling owners' profit of selling stocks soon afterwards. Instead of diverting existing corporate assets directly, the entrepreneurs siphon cash in the secondary market and from employees. However, the siphoned cash is not put to productive uses, as can be seen in the firms' deteriorating productivity and stock performance, as well as higher delisting risk, after ESOP adoption.

Leveraged ESOPs and ESOPs with high participation rates are associated with greater likelihood of controllers' cashing out and worse long-term firm performance. Controlling shareholders with tunneling motives are likely to cajole as many employees as possible to participate, sometimes with loans for leverage ESOPs, so as to inflate the stock prices for personal gains. As mentioned above, some controlling owners persuade employees to participate in ESOPs by promising to compensate them in case of any losses. Unfortunately, those promises are seldom kept. Although the CSRC prohibits compulsory ESOP participation, the lack of specific punishment gives entrepreneurs the incentive to manipulate and expropriate employees and public investors.

**Table 9**

Regression results of BHAR, DEA efficiency, TFP, and Tobin's Q on ESOP and firm characteristics: The whole sample. Model I-III is cross-sectional regression using ESOP firms. Model IV and V is fixed effects, and Model VI OLS, DID test using ESOP and PSM firms. The financial variables for Model I-III are the 3-year average before ESOP adoption announcements, and those for Model IV-VI are annual values. The variables are defined in Appendix A. Standard errors are in parenthesis. *, **, and ***denote significance at the 10%, 5%, and 1% levels, respectively.

| Model | I | II | III | IV | V | VI |
|---|---|---|---|---|---|---|
| Dependent variable | | BHAR | | DEA efficiency | TFP | Tobin's Q |
| **Accounting accruals** | **-0.007*****<br>**(0.002)** | | | | 0.029***<br>(0.006) | 0.684<br>(0.449) |
| **Private enterprise×ESOP_Leverage** | | **-0.343*****<br>**(0.123)** | | | | |
| **ESOP_participation×CC_separation** | | | **-0.009****<br>**(0.004)** | | | |
| **PostESOP×Accounting accruals_lag2** | | | | **-0.075**<br>**(0.062)** | | |
| **PostESOP×Accounting accruals** | | | | | **-0.160*****<br>**(0.058)** | **-0.646****<br>**(0.272)** |
| ESOP_Leverage | | 0.270**<br>(0.117) | | | | |
| Ownership category<br>(ref.= Local SOE) | | | | | | |
| Central SOE | | 0.224<br>(0.277) | | | | |
| Private enterprise | | -0.035<br>(0.109) | | | | |
| Public enterprise | | -0.096<br>(0.156) | | | | |
| ESOP_participation | | | 0.038<br>(0.040) | | | |
| CC_separation | | | 0.002<br>(0.003) | | | |
| PostESOP | | | | -0.009<br>(0.008) | -0.135**<br>(0.061) | 0.194**<br>(0.088) |
| Accounting accruals_lag2 | | | | -0.004<br>(0.005) | | |
| ESOP | | | | | | -0.528***<br>(0.053) |
| Post | | | | | | -0.400***<br>(0.150) |
| Log(sales) | 0.022<br>(0.028) | 0.022<br>(0.027) | 0.031<br>(0.031) | 0.050***<br>(0.006) | -0.133***<br>(0.042) | -0.179***<br>(0.035) |
| Leverage | -0.177<br>(0.172) | -0.209<br>(0.189) | -0.257<br>(0.180) | -0.008**<br>(0.004) | 0.029**<br>(0.012) | 2.095***<br>(0.575) |

| | | | | | | |
|---|---|---|---|---|---|---|
| R&D/Sales | -0.355 | -0.319 | -0.343 | -0.067*** | -0.603 | 1.181** |
| | (0.451) | (0.470) | (0.918) | (0.024) | (1.284) | (0.480) |
| Ad/Sales | -0.203 | -0.165 | -0.146 | -0.334*** | -10.437** | 0.785 |
| | (0.192) | (0.204) | (0.203) | (0.063) | (4.155) | (0.752) |
| Capital/Labor | 0.046 | 0.040 | 0.038 | -0.000 | 0.010*** | -0.016*** |
| | (0.043) | (0.048) | (0.045) | (0.002) | (0.003) | (0.005) |
| Log(age) | 0.049 | 0.030 | 0.014 | -0.104 | -0.414* | 0.528*** |
| | (0.063) | (0.063) | (0.064) | (0.070) | (0.251) | (0.118) |
| Intercept | -0.580 | -0.350 | -0.231 | 1.006* | 2.836*** | -2.144** |
| | (0.642) | (0.624) | (0.664) | (0.601) | (0.671) | (0.942) |
| Industry & year effects controlled | | | | | | |
| Observations | 541 | 540 | 505 | 10336 | 3761 | 14477 |
| R-squared | 0.034 | 0.045 | 0.051 | 0.060 | 0.067 | 0.121 |

5.1.2. Swindle public investors through information asymmetry

Because public investors are unlikely to purchase the firm's stocks at prices above fundamental value, if information is easily accessible. To understand how controlling shareholders in China successfully siphoned cash from public investors, we analyze the CAR, BHAR, and likelihood of equity sales of firms with small and large size, as information asymmetry is higher for small firms. Table 10 reports the results. CAR is negatively correlated with Log(size) at the 1% significance level. The smaller the market cap, the easier it is to inflate the firm's stock price through ESOP adoption announcement. When employees are no-so-numerous, the likelihood of equity sales is significantly positively correlated with CAR and BHAR is significantly negatively correlated with Accounting accruals, but the relationship is insignificant otherwise. Because most of the firms that designated ST/*ST or delisted within 3 years of ESOP adoption have not-so-numerous employees, we do not conduct similar tests for delisting risk.

We also conduct DID tests for firms with numerous and not-so-numerous employees separately, and report the results in Table 11. In the test for DEA efficiency and TFP, the coefficient of PostESOP is significantly negative when employees are not-so-numerous, but insignificant when employees are numerous. The relative change in Tobin's Q from ESOP adoption is also significantly worse in the group with not-so-numerous employees.

The results suggest that tunneling through ESOPs is concentrated in firms with not-too-many employees, because smaller firms have higher information asymmetry and their stock prices can be manipulated more easily. For the whole sample, the effect of tunneling due to weak legal institutions outweighs the free-rider effect.

**Table 10**

Regression (cross-sectional) results of CAR, equity sales and BHAR on firm characteristics using ESOP firms. Model I is for the whole sample and its dependent variable is CAR (-5,5). For Model II-V, the whole sample is divided into 2 sets of subsamples based on the average number of employees for 3 years before ESOP adoption announcements. Numerous-employee firms are those in the top quartile in terms of the number of employees. The dependent variable for logistic regression II and III equals 1 if the controlling shareholder sells the company's stocks within 1 month of the ESOP adoption announcement and 0 otherwise. All financial variables are the 3-year average before ESOP adoption announcements. The explanatory variable CAR is CAR (-10,30). The variables are defined in Appendix A. Standard errors are in parenthesis. *, **, and ***denote significance at the 10%, 5%, and 1% levels, respectively.

| Model | I | II | III | IV | V |
|---|---|---|---|---|---|
| Dependent variable | CAR | Equity sales | | BHAR | |
| Employees | | Numerous | Not-so-numerous | Numerous | Not-so-numerous |
| **Log(size)** | **-0.037*** ** | | | | |
| | **(0.011)** | | | | |
| **CAR** | | **0.324** | **1.611** ** | | |
| | | **(2.109)** | **(0.636)** | | |
| **Accounting accruals** | | | | **1.660** | **-0.006*** ** |
| | | | | **(2.042)** | **(0.002)** |
| Log(sales) | 0.016** | 0.762** | 0.233 | 0.038 | -0.043** |
| | (0.007) | (0.332) | (0.179) | (0.072) | (0.020) |
| Leverage | -0.009 | -3.312* | -0.910 | -0.558 | -0.007 |
| | (0.036) | (1.949) | (0.967) | (0.541) | (0.113) |
| R&D/Sales | -0.337 | -164.966* | 5.005 | 6.822 | -0.452 |
| | (0.235) | (99.800) | (3.719) | (6.633) | (0.473) |
| Ad/Sales | 0.165** | 5.706** | 0.372 | -0.298 | -0.360* |
| | (0.067) | (2.847) | (1.503) | (0.472) | (0.213) |
| Capital/Labor | 0.021 | -0.210 | -0.626 | 0.278** | 0.009 |
| | (0.015) | (0.932) | (0.499) | (0.139) | (0.047) |
| Log(age) | 0.011 | -0.965 | -0.574 | 0.224 | -0.033 |
| | (0.018) | (0.999) | (0.461) | (0.155) | (0.069) |
| Intercept | 0.057 | 1.145 | 2.593 | -2.680* | 0.670 |
| | (0.179) | (8.611) | (4.173) | (1.457) | (0.695) |
| Industry & year effects controlled | | | | | |
| Observations | 541 | 125 | 384 | 135 | 406 |
| R-squared | 0.078 | | | 0.132 | 0.059 |
| Pseudo $R^2$ | | 0.181 | 0.056 | | |

**Table 11**

Regression (OLS) results of firm productivity on firm characteristics using ESOP and PSM firms: The subsamples. The whole sample is divided into 2 sets of subsamples based on the average number of employees for 3 years before ESOP adoption announcements. Numerous-employee firms are those in the top quartile in terms of the number of employees. The variables are defined in Appendix A. Standard errors are in parenthesis. *, **, and ***denote significance at the 10%, 5%, and 1% levels, respectively.

| Model | I | II | III | IV | V | VI |
|---|---|---|---|---|---|---|
| Dependent variable | DEA efficiency | | TFP | | Tobin's Q | |
| Employees | Numerous | Not-so-numerous | Numerous | Not-so-numerous | Numerous | Not-so-numerous |
| **PostESOP** | **-0.012** | **-0.016**** | **-0.136** | **-0.148**** | **0.129**** | **0.143** |
| | **(0.010)** | **(0.007)** | **(0.088)** | **(0.068)** | **(0.063)** | **(0.111)** |
| ESOP | 0.031*** | 0.006 | 0.341*** | 0.108** | -0.182*** | -0.584*** |
| | (0.006) | (0.004) | (0.061) | (0.050) | (0.040) | (0.063) |
| Post | -0.004 | -0.020*** | 0.020 | -0.112** | -0.299*** | -0.427** |
| | (0.008) | (0.005) | (0.066) | (0.053) | (0.073) | (0.194) |
| Log(size) | 0.027*** | 0.054*** | -0.319*** | -0.157*** | 0.149*** | 0.080* |
| | (0.003) | (0.003) | (0.023) | (0.032) | (0.023) | (0.045) |
| Leverage | -0.003 | -0.013* | 0.063*** | 0.021*** | 1.030* | 1.745*** |
| | (0.030) | (0.007) | (0.021) | (0.008) | (0.589) | (0.441) |
| R&D/Sales | -0.036 | -0.192*** | -2.043* | -1.466 | 1.583 | 0.777* |
| | (0.117) | (0.028) | (1.114) | (0.909) | (1.540) | (0.410) |
| Ad/Sales | -0.047*** | -0.026*** | -10.890** | -9.151** | 4.504*** | 0.271 |
| | (0.013) | (0.009) | (4.336) | (3.827) | (0.395) | (0.245) |
| Capital/Labor | -0.010* | -0.003*** | -0.100** | 0.005*** | -0.055* | -0.020*** |
| | (0.005) | (0.001) | (0.045) | (0.001) | (0.031) | (0.006) |
| Log(age) | 0.006 | -0.009* | 0.195*** | 0.172*** | -0.027 | 0.948*** |
| | (0.007) | (0.005) | (0.051) | (0.058) | (0.063) | (0.174) |
| Intercept | 0.313*** | 0.257*** | 2.315*** | 1.598*** | -0.708 | -7.464*** |
| | (0.068) | (0.049) | (0.297) | (0.290) | (0.502) | (1.382) |
| Industry & year effects controlled | | | | | | |
| Observations | 3719 | 10756 | 788 | 2973 | 3719 | 10758 |
| R-squared | 0.282 | 0.200 | 0.415 | 0.049 | 0.550 | 0.129 |

### 5.1.3. Misleading employees when market sentiment is high

Why do the employees buy the ESOP shares if the controlling shareholder intends to tunnel through ESOPs? Because employees have inside information about the firm, it is difficult for the controlling owner to fool them about firm value. However, some employees are likely to neglect fundamentals when market sentiment is high. In addition, if the controlling shareholder manages earnings upward before ESOP adoption and offer loans to the employees, the good earnings, market optimism, and readily available financing are likely to make employees purchase ESOP shares rather than other assets.

Following Baker & Wurgler (2006), we form an index to measure investor sentiment. As mentioned by Zhu & Niu (2016), the number of IPOs, the percentage of equity financing, and dividend premium are not suitable for measuring investor sentiment in China, due to the regulation on equity financing and the low dividend level. Due to the price restriction for stocks in A share market, we do not use the average first-day return on IPO. We use 5 variables to reflect the level of investor sentiment for A-share market: the close-end fund discount, A-share turnover rate, consumer confidence index, the number of new A-share market accounts, and the average first-day turnover on IPO. Following Baker & Wurgler (2006), we regress the 5 variables on macroeconomic indicators from January, 2014 to December, 2018, and use the residual as sentiment proxies, whose first principal component is the monthly sentiment index. The macroeconomic variables include industrial added value, macroeconomic climate index, CPI, growth rate, money supply M1, real GDP growth rate, and unemployment rate. Because the National Bureau of Statistics of China only reports seasonal data for unemployment rate and GDP growth, we estimate their monthly value by assuming a constant growth rate within each season. The number of IPOs is zero for 6 months, for which we use the remaining 4 sentiment proxies to calculate the sentiment index.

We use investor sentiment of the month of ESOP adoption announcement in the logistic regression for delisting risk. The interactive effects of investor sentiment shed light on how tunneling controllers successfully swindle employees. In Table 12, Sentiment×Equity sales is significantly positively correlated with delisting risk when Accounting accruals or Leverage is high, but insignificant otherwise. The coefficient of Sentiment×Other receivables/assets is significantly positive only in the group with high CC_separation, or the group with low Salary_Nonmgt.

Controllers with incentivization motives are likely to be indifferent between announcing ESOP adoption in a high- or low-sentiment month. However, controllers with tunneling motives are much more likely to announce ESOP adoption in a high-sentiment month, because employees as insiders are unlikely to fall into their trap when sentiment is low. Using the likelihood of ESOP adoption as the dependent variable, we find that firms at higher risk of tunneling are more likely to announce ESOP adoption when market sentiment is high. For instance, the interaction term Sentiment × Accounting accruals is significantly positively associated with the likelihood of ESOP adoption. To adjust for pre-existing conditions, we also conduct DID tests on the subsample with high and not-so-high sentiment, the former of which is in the top quartile in terms of investor sentiment in the month of ESOP adoption announcements. The results are consistent with our previous findings: the relative change in productivity and Tobin's Q is worse if ESOP adoption is announced in a high-sentiment month[1].

**Table 12**

Logistic regression (cross-sectional) results of delisting risk on firm characteristics using ESOP and PSM firms: The subsamples. The whole sample is divided into 2 sets of subsamples based on the average Accounting accruals, Leverage, Salary_Nonmgt, and CC_separation for 3 years before ESOP adoption announcements. The level "High" refers to being more than the median of the grouping criteria in the second row, and "Low" the opposite. The dependent variable equals 1 if the firm is designated ST/*ST or delisted within 3 years of ESOP adoption, and 0 otherwise. All financial variables are the 3-year average before ESOP adoption announcements. The variables are defined in Appendix A. Standard errors are in parenthesis. *, **, and ***denote significance at the 10%, 5%, and 1% levels, respectively.

| Model | I | II | III | IV | V | VI | VII | VIII |
|---|---|---|---|---|---|---|---|---|
| Grouping criteria | Accounting accruals | | Leverage | | Salary_Nonmgt | | CC_separation | |
| Level | High | Low | High | Low | High | Low | High | Low |
| **Sentiment×Equity sales** | **1.004*** ** | **0.293** | **0.826*** | **-0.223** | | | | |
| | **(0.235)** | **(0.295)** | **(0.438)** | **(0.158)** | | | | |
| **Sentiment×Other receivables/assets** | | | | | **2.862** | **3.077** ** | **4.773** ** | **-2.045** |
| | | | | | **(6.950)** | **(1.346)** | **(2.107)** | **(4.199)** |
| Sentiment | -0.026 | -0.037 | -0.043 | -0.000 | -0.065 | -0.086 | -0.033 | -0.070 |
| | (0.090) | (0.110) | (0.085) | (0.109) | (0.123) | (0.103) | (0.090) | (0.127) |
| Equity sales | -3.288*** | 0.371 | -1.099 | -0.262 | | | | |
| | (0.931) | (0.781) | (1.271) | (1.019) | | | | |
| Other receivables/assets | | | | | 4.345 | 1.087 | 3.023 | 3.339 |

[1] Table available upon request.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | (3.724) | (1.782) | (2.064) | (3.165) |
| Log(sales) | -0.319*** | -0.540*** | -0.354*** | -0.815*** | -0.479*** | -0.408*** | -0.405*** | -0.480*** |
| | (0.109) | (0.109) | (0.075) | (0.236) | (0.110) | (0.093) | (0.090) | (0.110) |
| Leverage | 3.671*** | 5.230*** | 3.846*** | 5.914** | 3.758*** | 5.503*** | 3.588*** | 5.553*** |
| | (0.694) | (0.775) | (0.836) | (2.325) | (0.849) | (0.749) | (0.712) | (0.878) |
| R&D/Sales | 2.958*** | 0.331 | 3.493 | 3.146*** | 4.353 | 2.674*** | 2.691*** | -3.618 |
| | (0.911) | (6.933) | (4.738) | (0.756) | (2.788) | (0.666) | (0.706) | (8.310) |
| Ad/Sales | -0.332 | -0.296 | -0.381 | -0.238 | 2.830 | -2.185 | 1.111 | -2.949* |
| | (1.871) | (1.654) | (1.423) | (2.220) | (1.758) | (2.037) | (1.712) | (1.722) |
| Capital/Labor | -0.133 | -0.322 | -0.068 | -1.833** | -0.236 | -0.103 | -0.280 | -0.131 |
| | (0.149) | (0.238) | (0.111) | (0.819) | (0.161) | (0.139) | (0.187) | (0.188) |
| Log(age) | -0.040 | -0.185 | -0.380 | 0.589 | 0.156 | -0.121 | 0.178 | -0.135 |
| | (0.441) | (0.461) | (0.376) | (0.592) | (0.403) | (0.448) | (0.381) | (0.491) |
| Intercept | -1.215 | -1.166 | 1.215 | -3.794 | -2.984 | -1.017 | -3.297 | -0.970 |
| | (3.976) | (4.340) | (3.381) | (5.755) | (3.746) | (4.016) | (3.467) | (4.553) |
| Industry & year effects controlled | | | | | | | | |
| Observations | 899 | 847 | 855 | 891 | 995 | 1012 | 1017 | 998 |
| Pseudo $R^2$ | 0.119 | 0.168 | 0.092 | 0.166 | 0.096 | 0.224 | 0.140 | 0.201 |

### 5.2. How much do they tunnel?

To estimate the magnitude of tunneling, we examine the abnormal losses of investors from ESOP adoption announcements to lockup expiration. The long-term abnormal return is not significantly different from zero for the whole sample. Then we separate it into subsamples based on the top and bottom quartile in terms of the grouping criteria in the leftmost column of Table 13. BHAR is statistically significant with a mean of -11.8% when Other receivables/assets and Accounting accruals are both high before ESOP adoption announcements, but insignificant otherwise. BHAR is significant with a mean of -5.3% when the ESOP is leveraged, but insignificant otherwise. This pattern continued for the combined criteria of Non-recur income/profits and Accounting accruals.

To derive the abnormal change in market cap, we multiply the initial market cap by BHAR for each firm. The total abnormal change in market cap is -81.19 billion RMB for the group of sample firms with high other receivables/assets and Accounting accruals, and it is -82.83 billion RMB for the group of firms using leveraged ESOPs. The abnormal reduction in market cap indicates that tunneling through ESOPs is economically significant in China.

**Table 13**

Summary of BHAR for sample firms. The level "high" means being in the top quartile and "low" the bottom quartile in terms of the grouping criteria in the leftmost column. *, **, and ***denote significance at the 10%, 5%, and 1% levels, respectively.

| Subgroup | Number of firms | Mean | t-statistic | Percent positive | Wilcoxon signed-rank z-statistic |
|---|---|---|---|---|---|
| High Other receivable/assets & high Accounting accruals | 48 | -11.8%*** | -3.28 | 29.2% | -3.33 |
| Low Other receivable/assets & low Accounting accruals | 43 | 0.6% | 0.12 | 46.5% | 0.06 |
| ESOP_leverage=1 | 132 | -5.3%* | -1.77 | 36.4% | -3.01 |
| ESOP_leverage=0 | 426 | 2.0% | 0.95 | 45.1% | -1.58 |
| High Non-recur income/profits & high Accounting accruals | 30 | -8.1%* | -1.94 | 30.0% | -2.09 |
| Low Non-recur income/profits & low Accounting accruals | 39 | -0.8% | -0.15 | 48.7% | -0.42 |

### 5.3. Why do they tunnel?

From our results above, the higher leverage and consequently higher bankruptcy risk do not discipline the owner-managers but rather are associated with more tunneling in China. Friedman, Johnson, and Mitton (2003) argued that debt may act as a credible commitment for entrepreneurs to prop up their firms under mild adverse shocks in counties with weak investor protection. However, entrepreneurs' short-term propping is for long-term tunneling, and their intention to transfer wealth from minority shareholders should be reflected in the initial valuation of their highly levered firms under classical finance assumptions. The initial overperformance and subsequent underperformance of firms subjected to tunneling suggest that investors cannot directly observe tunneling when it is going on.

Why are controlling owners more likely to tunnel when their firms have higher leverage or more previous tunneling? Entrepreneurs have the choice to invest in positive-NPV projects or to tunnel, and they probably will have tunneling motives if their net return from tunneling ($R_{tunneling}$) greatly exceeds and their net return from investing ($R_{investing}$). The lack of punishment on entrepreneurs results in extremely high $R_{tunneling}$ in China. According to CSRC regulations, legal punishment for controlling shareholders who tunneled billions of RMB from investors is a small fine (less than 1 million RMB) plus restriction in future IPO. For example, a high-tech imaging firm, Protruly Vision (600074.SH) adopted ESOP in 2015 and have an average debt ratio of 71.6% for 3 years before ESOP adoption announcement. By 2017, the controller of Protruly Vision tunneled 6.7 billion RMB from the listed firm and paid less than 0.6 million RMB in fines to the CSRC. For $R_{investing}$ to exceed the high $R_{tunneling}$ in China, the entrepreneur most likely needs skills to invest in high-risk projects.

$R_{tunneling}$ most likely greatly exceed $R_{investing}$ for highly levered firms in countries of weak investor protection. Consider two firms with similar $R_{tunneling}$: Firm A, which has high leverage, and Firm B, which has low leverage. Firm A most likely mainly invests in low-risk projects, because firms are unable to support high interest and principle payment using cash flow from their high-risk projects. Firm B, on the other hand, is free to invest in positive-NPV projects with high cash-flow volatility, which will yield higher $R_{investing}$. So $R_{tunneling}$ is more likely to exceed $R_{investing}$ for Firm A than for Firm B. The entrepreneur in Firm A may mislead investors into believing the cash is invested in high-tech projects, as in the case of Protruly Vision.

$R_{tunneling}$ is also likely to greatly exceed $R_{investing}$ for firms subjected to severe tunneling in the past. Because the current punishment on entrepreneurs who tunnel billions of RMB is still a small fine of less than 1 million RMB, $R_{tunneling}$ is almost unchanged in China. Even if the firm invests in high-risk projects, their NPV are likely to be small because the entrepreneur has already tunneled a large amount of corporate assets from the projects. Besides, the entrepreneur most likely lacks the management skills for investing in high-growth projects, so ($R_{tunneling}$-$R_{investing}$) is still likely to be positive and large. Entrepreneurs who lack management abilities or resources to invest in high-return projects may obtain equity financing through IPO, then issue excess debt, and finally divert the cash obtained through intercorporate loans, related-party transactions, and new channels such as ESOPs.

Why is higher salary level negatively associated with tunneling? Firms in high-growth industries are more likely to be labor-intensive, rather than capital-intensive, so firms paying higher salary to non-management employees tend to have higher $R_{investing}$, which reduces the motive for tunneling.

Similar to Atanasov et al. (2010), our findings, together with previous research on tunneling, suggest that market participants can adapt their strategies for maximizing private benefits to regulatory change. In China, controllers of listed firms have increased their tunneling through less detectable channels after regulatory changes in 2005 and 2006 made it more difficult to tunnel through non-operational funds. For example, the owner of Kangdexin (stock code 002450) inflated profits by 11.9 billion RMB from 2015 to 2018 and tunneled 12.2 billion BMB from the listed firm through related-party transactions. Besides operational tunneling, corporate controllers also started to disguise their financial tunneling as value-adding events after 2006. The controlling shareholders and market regulators tend to coevolve and regulatory tightening may only curb tunneling in the short run as long as the root causes of the agency problem between controlling and minority shareholders are unchanged. The lack of legal punishment on the offenders may be one of the root causes of persistent tunneling in the Chinese market. For instance, the controller of Kangdexin only paid 0.6 million RMB of fines to the CSRC, in stark contrast to the severity of the legal punishment in the Enron Scandal.

Looking ahead, we note that the new regulations in China may not fully prevent the insider abuse identified in this paper. Although the New Asset Management Rules limit the leverage of financial products, controlling shareholders can still increase ESOP leverage by lending funds to employees before assembling tranche products after the regulatory change. The CSRC also issued the New Stock Sale Rules in 2017, which prevent large shareholders (controlling >5% of total shares) from selling more than 1% of the firm's shares in the secondary market within 90 days. However, the controlling shareholders can sell shares in block to private firms or individuals in advance (6 months earlier) and then share the profit with them after they cash out in the secondary market on ESOP adoption announcement. The Shanghai and Shenzhen Stock Exchange issued new guidelines on ESOPs in 2019, which stated that firm controllers should not trade their firms' stocks during ESOP adoption. However, there is no specific punishment on controllers who violate the guidelines. Increased legal punishment may be necessary to reduce tunneling in emerging markets.

**6. Robustness tests**

Besides productivity, shareholder value, and delisting risk, we also measure firm performance in terms of operating cash flow. To measure operating performance, we divided each sample firm's operating cash flow (OCF) by its total assets (TA) for each of the 3 years before and after the ESOP adoption announcement, which we called the unadjusted OCF/TA. Then we calculated the adjusted OCF/TA by subtracting the control group's OCF/TA from the unadjusted OCF/TA for each year. We used two types of controls: the industry average according to Wind Level 1 Industry Classification, and the PSM control average. We define "marginal operating performance" as the difference between the average PSM-control-adjusted OCF/TA for 2 years after and for 2 years before ESOP adoption announcements, and "cumulative operating performance" as the sum of PSM-control-adjusted OCF/TA for 3 years after ESOP adoption announcements.

The industry-average-adjusted OCF/TA of sample firms was indistinguishable from that of benchmark before ESOP adoption, except for being worse than that of benchmark 1 year before ESOP adoption at the 10% significance level. However, the operating performance of sample firms considerably deteriorated afterwards, being 1.4% and 1.3% lower than that of benchmark for 2 and 3 years after ESOP adoption, respectively, both at the 1% significance level. The PSM results were similar: the operating performance of sample firms was not significantly different from that of benchmark before ESOP adoption, but was worse for 2 and 3 years after ESOP adoption, both at the 5% level. The 3-year post-ESOP cumulative OCF/TA was -3.9% at the 1% significance level for the industry-adjusted approach, and -1.6% at the 10% level for the PSM approach.[2]

We perform similar cross-sectional regression for operating performance. Salary_Nonmgt is significantly positive at the 1% level, supporting incentivization motives. CAR×CC_separation, Other receivables/assets, and Leverage are significantly negatively associated with operating performance, supporting tunneling motives. ESOP_leverage, Non-recur income/profits×Log(ESOP_participants), and ESOP_participation are all significantly negative, supporting tunneling motives for controllers using leveraged ESOPs and ESOPs with high participation rates in Chinese listed firms.

To address endogeneity concern, we use PSM-DID approach to alleviate sample selection bias and omitted variable bias. We first conduct a DID test on operating performance. In both fixed-effects and OLS regressions using ESOP and PSM firms, PostESOP is significantly negative. And its interaction with firm characteristics show that operating performance deteriorates after ESOP adoption for firms with high Accounting accruals, Leverage, Other receivables/assets or CC_separation.

Then, we include PSM control firms in regressions on the likelihood of controlling owners' immediate equity sales. In pooled logistic regressions, CAR has positive coefficients at the 1% level, and Accounting accruals_lag1×ESOP_participation×CAR has a positively significant coefficient at the 1% level. Consistent with our previous findings, the following variables are positively correlated with the likelihood of controllers' equity sales: ESOP_leverage×ESOP_participation, Other receivables/assets×Non-recur income/profits_lag2, and ESOP_leverage×Leverage.

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[2] The tables in Section 6 are available upon requests.

Finally, we include ESOP and PSM firms for the test on delisting risk and find consistent results. Salary/employee_nonmgt is negatively associated with delisting risk. Accounting accruals × CC_separation is positively correlated with delisting risk at the 5% level. The coefficient of ESOP× CC_separation is positive, but insignificant. The results for subsamples show that Accounting accruals × CC_separation is positive at the 1% level when Other receivables/assets is high, while insignificant otherwise. ESOP× Leverage is significantly positive when Other receivables/assets is high, while insignificant otherwise.

Our main results are also robust to changes in variable specifications. We replace the end of the holding period for BHAR as 1 year after the unlock date. The regressions using BHAR with an extended holding period show that these variables are significantly negatively associated with BHAR: CC_separation×ESOP_participation, Leverage×Non-recur income/profits, Private enterprise×ESOP_leverage. Salary_Nonmgt is significantly positive at the 1% level.

We change the proxy of earnings management to abnormal accruals, using the cross-sectional version of Jones model to estimate discretionary current accruals (Jones, 1991; Dechow et al., 1995; Teoh et al., 1998). The details of the procedure are in Appendix B. Regressions using Abnormal accruals yield similar results as above. For example, Abnormal accruals is negatively correlated with operating performance at the 5% level, while the interaction of Abnormal accruals, as well as its lagged value, with ESOP_participation and CAR is positively associated with the likelihood of controllers' equity sales.

In cross-sectional regressions, we replace ESOP_participation with Log(ESOP_participants), and Salary_Nonmgt with Salary/employee_nonmgt. We also add Ownership category and Book to market to the control variables. We use both the market and market-adjusted model in CAR calculation, and replace CAR (-5,5) with CAR (-3,3) and CAR (-8,8) for the empirical tests. We change the number of PSM firms from 5 to 10 and 30 for each ESOP firm, and use operating performance derived from those PSM firms in cross-sectional regressions. To address reverse causality concerns, we replace Accounting accruals and Leverage with their lagged values in fixed-effects models. For the DID tests, we change fixed effects to OLS regression, and vice versa. The results are qualitatively similar for the changes in variable specifications or regression method.

## 7. Conclusions

This paper investigates the motives of ESOP adoption for Chinese listed firms during the period 2014 to 2018. Firms with higher salary level are more likely to adopt ESOPs to incentivize employees. However, controlling shareholders with tunneling motives disguise the adoption of ESOPs as a value-adding event, and their firms have higher leverage, more intercorporate loans, and greater separation of ownership and control. These controllers inflate the stock prices through ESOP adoption announcements, and then cash out soon afterwards. For their firms, the short-term increase in stock prices is not translated into long-term value for minority investors, and their market cap shrinks, productivity decreases, and delisting risk increases after ESOP adoption. This paper identifies several tactics that tunneling controllers use to inflate stock prices in the short term: earnings management, leveraged ESOPs, and ESOPs with high participation rates. When legal protection for investors is weak, controlling owners of listed firms tend to obtain excess debt and tunnel through intercorporate loans and new channels such as ESOPs.

These findings have implications for investors, regulators, and researchers. Investors may want to discriminate among entrepreneurs who adopt ESOPs to motivate employees and those who use ESOPs to tunnel. Regulators in countries with high ownership concentration may find these results useful for preventing insider abuse in ESOP adoption. Limiting the number of ESOP participants and the use of leveraged ESOPs may reduce tunneling through ESOPs. Regulators and market participants tend to coevolve, and regulators need to continuously monitor the effectiveness of existing policies and update regulatory strategies when necessary. Researchers studying ESOPs in emerging markets may need to consider tunneling as a possible motivation. The coevolution between regulators and market participants may necessitate researchers to consider other forms of tunneling when evaluating new regulations for reducing one form of tunneling. While this study only tests equity sales as a way to cash out, there is also anecdotal evidence that tunneling controllers cash out by pledging stocks soon after ESOP adoption announcements. And Chinese state-owned firms face restrictions in equity sales, so their controlling owners could tunnel through ESOPs in ways not identified in this paper.

## Competing Interest Statement

Declarations of interest: none.


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# Appendix A

**Table A1**

Definitions of Variables

| Variable Name | Definition |
| --- | --- |
| Other receivables/assets | 3-year average of other receivables divided by total assets prior to ESOP announcements. |
| Accounting accruals | 3-year average of (net income-operating cash flow)/(total assets) prior to ESOP announcements. |
| ESOP_leverage | Dummy variable equal to 1 if the ESOP uses loans from controlling shareholders or financial institutions, and 0 otherwise. |
| Non-recur income/profits | 3-year average of non-recurring income divided by net profit prior to ESOP announcements. |
| ESOP_participation | Number of ESOP participants divided by the number of employees at the firm. |
| Ownership category | Categorical variable equal to 1 for local state-owned enterprise, 2 for central SOE, 3 for private enterprise, and 4 for public enterprise. |
| Log(size) | Natural logarithm of 3-year average of market capitalization prior to ESOP announcements. |
| Leverage | 3-year average of total liabilities divided by total assets prior to ESOP announcements. |
| Salary_Nonmgt | 3-year average of non-management employees' salary divided by net income prior to ESOP announcements, where non-management employees exclude board members and upper managers. |
| Book to market | 3-year average of equity book value divided by market capitalization prior to ESOP announcements. |
| Industry | Categorical variable equal to 1 for healthcare, 2 for consumer staples, 3 for real estate, 4 for IT, 5 for manufacturing, 6 for consumer discretionary, 7 for materials, 8 for utilities, 9 for energy, and 10 for telecom. |
| CC_separation | 3-year average of separation of control and cash-flow rights prior to ESOP announcements. |
| Log(ESOP_participants) | Natural logarithm of the number of ESOP participants. |
| PostESOP | Dummy variable equal to 1 for the observations of ESOP firms after ESOP adoption announcements; it is equal to 0 for all the observations of PSM firms and for the observations of ESOP firms on or before ESOP adoption announcement day. |
| Post | Dummy variable equal to 1 if the current year is after ESOP adoption announcement and to 0 otherwise, where the ESOP adoption announcement date for companies in the control group is the same as that of the ESOP-adopting firm with the closest propensity score. |
| ESOP | Dummy variable equal to 1 for ESOP-adopting firms and to 0 otherwise. |
| Year | The year of ESOP adoption announcements. |
| Sentiment | Investor sentiment index for the month of ESOP adoption announcements, calculated in Section 5.1.3. |
| Equity sales | Dummy variable equal to 1 if the controller sells the firm's stocks within 1 month of announcing ESOP adoption, and 0 otherwise. |
| Salary/employee_nonmgt | The total salary for non-management employees divided by the number of non-management employees. |
| Log(sales) | The natural logarithm of sales revenue. |
| R&D/Sales | R&D expenditure divided by sales revenue. |
| Ad/Sales | Advertising expenditure divided by sales revenue. |
| Capital/Labor | Inflation adjusted net PP&E divided by the number of employees. |
| Log(age) | The natural logarithm of the number of years since the firm was established. |

# Appendix B: Calculation of Abnormal Accruals

Following Dechow et al. (1995) and Teoh et al. (1998), we calculate abnormal accruals as the residual in current accruals after deducting nondiscretionary current accruals. As specified by Equation (5), we first run cross-sectional OLS regression on PSM control firms for each of the 9 industries and each year from 2011 to 2018:

$$\frac{CA_{i,t}}{TA_{i,t-1}} = a\frac{1}{TA_{i,t-1}} + b\frac{\Delta Sales_{i,t}}{TA_{i,t-1}} + \varepsilon_{i,t} \quad (5)$$

Where $TA_{i,t-1}$ is the total assets for firm i in year $t-1$; $\Delta Sales_{i,t}$ is the change in sales revenue for firm i in year t; $CA_{i,t}$ is current accruals for firm i in year t, calculated as:

$$CA_{i,t} = \Delta noncash\ current\ assets_{i,t} - \Delta nondebt\ current\ liabilities_{i,t} \quad (6)$$

Then we use the estimated regression coefficients for each industry-year combination from Eq. (5) to calculate each sample firm's nondiscretionary current accruals (NDCA) as follows:

$$NDCA_{i,t} = \hat{a}\frac{1}{TA_{i,t-1}} + \hat{b}\frac{\Delta Sales_{i,t} - \Delta TR_{i,t}}{TA_{i,t-1}} \quad (7)$$

Where $\hat{a}$ and $\hat{b}$ are the estimated regression coefficients in Eq. (5) and $\Delta TR_{i,t}$ is the change in trade receivables for firm i in year t.

Finally, abnormal accruals for each sample firm is as follows:

$$Abnormal\ accruals_{i,t} = \frac{CA_{i,t}}{TA_{i,t-1}} - NDCA_{i,t} \quad (8)$$

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